

Proof. Let $X' : \Omega \rightarrow \mathbb{R}^d$ be any other random data vector with finite first moment.

Since all constant predictors exist in our predictor class $\mathcal{V} = \{\eta(\cdot; \theta)\}$, the best loss achievable on (X', Z) by functions in $\mathcal{V}$ must be at least as good as the trivially attainable loss (the best loss available by such constant predictors):

$$\inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X'; \theta), Z)] \leq L_{\tau}$$

By assumption, the trivially attainable loss cannot be improved upon over (X, Z) by predictors in $\mathcal{V}$:

$$L_{\tau} = \inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X; \theta), Z)]$$

Since our choice of X' was arbitrary, this shows that X maximizes the minimal achievable loss, so X $(\mathcal{V}, \mathcal{L})$ -guards Z . $\square$

C Linear Guardedness is Equivalent to Linear Statistical Parity

To measure the effect of linear guardedness on main-task classifiers, we use the following minimal definition of “fairness” with respect to an attribute, adapted from Edwards and Storkey [8].

Definition C.1 (Statistical Parity). *Let X and Z be defined as above, and let f be a function with domain $\mathbb{R}^d$. Then f exhibits **statistical parity** with respect to Z when evaluated on X if*

$$\forall z \in \mathcal{Z} : \mathbb{E}[f(X)|Z = z] = \mathbb{E}[f(X)].$$

We now prove the equivalence of conditions 3 and 5.

Theorem C.2. *Let X and Z be defined as above. Then every linear predictor $f(\mathbf{x}) = \mathbf{b} + \mathbf{W}\mathbf{x}$ exhibits statistical parity w.r.t. Z when evaluated on X if and only if each class-conditional mean $\mathbb{E}[X|Z = z]$ is equal to $\mathbb{E}[X]$.*

Proof. Suppose each class-conditional mean $\mathbb{E}[X|Z = z]$ is equal to $\mathbb{E}[X]$. Then by the linearity of expectation, we have for all $z \in \mathcal{Z}$:

$$\mathbb{E}[f(X)|Z = z] = \mathbb{E}[\mathbf{W}X + \mathbf{b}|Z = z] = \mathbf{W}\mathbb{E}[X|Z = z] + \mathbf{b} = \mathbf{W}\mathbb{E}[X] + \mathbf{b} = \mathbb{E}[f(X)].$$

This matches the definition of statistical parity provided in Definition C.1.

Conversely, suppose every linear predictor $f(\mathbf{x}) = \mathbf{b} + \mathbf{W}\mathbf{x}$ exhibits statistical parity w.r.t. Z when evaluated on X . Then this holds for the identity function $\text{id}(\mathbf{x}) = \mathbf{x}$, and thus for all $z \in \mathcal{Z}$:

$$\mathbb{E}[X|Z = z] = \mathbb{E}[\text{id}(X)|Z = z] = \mathbb{E}[\text{id}(X)] = \mathbb{E}[X].$$

$\square$

D Implications for Prior Work

In this section we discuss the implications of Theorem 4.1, which characterizes the necessary and sufficient conditions for an affine erasure function to yield a perfectly linearly guarded dataset, for methods proposed in prior work.

Spectral Attribute Removal (SAL) [36] uses the top n left singular vectors of Σ_{XZ} to construct an orthogonal projection matrix $\mathbf{Q}_{\text{SAL}} = \mathbf{I} - \mathbf{U}_{:n} \mathbf{U}_{:n}^T$ which is then applied to X . Notably, while n is presented as a free parameter in their method, all of their experiments involve binary classification problems where Z is a one-hot vector, and n is set to a value no greater than 2. We’ll call the version of SAL where $n = \text{rank}(\Sigma_{XZ})$, “full-rank SAL.” Since these left singular vectors are an orthonormal basis for Σ_{XZ} ’s column space, Theorem 4.1 implies that full-rank SAL guarantees linear guardedness.

Mean Projection (MP) [17] orthogonally projects X onto the orthogonal complement of the span of the difference in class centroids $\mathbb{E}[X|Z = 1] - \mathbb{E}[X|Z = 0]$, where Z is assumed to be binary. Since the centroids are equal after the projection, this method guarantees linear guardedness by Theorem 3.1. In fact, by Theorem 3.4, MP is mathematically equivalent to SAL when Z is a one-dimensional random vector taking one of two possible values.

E Derivation of LEACE

Theorem 4.2. Let X and Z be centered random vectors taking values in $\mathbb{R}^d$ and $\mathbb{R}^k$ respectively, each of finite second moment. Let $\mathbf{M} \in \mathbb{R}^{d \times d}$ be a p.s.d. matrix defining a (possibly degenerate) inner product on $\mathbb{R}^d$: $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{M}} = \mathbf{x}^T \mathbf{M} \mathbf{y}$. Let $\Sigma_{XX} \in \mathbb{R}^{d \times d}$ be X 's covariance matrix, and $\Sigma_{XZ} \in \mathbb{R}^{d \times k}$ be the cross-covariance matrix of X and Z . Let $\mathbf{A}^+$ denote the Moore-Penrose pseudoinverse of a matrix $\mathbf{A}$, and let $\mathbf{A}^{1/2}$ be the p.s.d. square root of a p.s.d. matrix $\mathbf{A}$. Then the objective

$$\operatorname{argmin}_{\mathbf{P} \in \mathbb{R}^{d \times d}} \mathbb{E} \left[\|\mathbf{P}X - X\|_{\mathbf{M}}^2 \right] \quad \text{subject to } \operatorname{Cov}(\mathbf{P}X, Z) = \mathbf{0}$$

has the following solution:

$$\mathbf{P}^* = \mathbf{I} - \mathbf{W}^+ \mathbf{P}_{\mathbf{W}\Sigma_{XZ}} \mathbf{W},$$

where $\mathbf{W}$ is the whitening transformation $(\Sigma_{XX}^{1/2})^+$ and $\mathbf{P}_{\mathbf{W}\Sigma_{XZ}} = (\mathbf{W}\Sigma_{XZ})(\mathbf{W}\Sigma_{XZ})^+$ is the orthogonal projection matrix onto $\operatorname{colsp}(\mathbf{W}\Sigma_{XZ})$.

Below are two independent proofs of Theorem 4.2.

E.1 Algebraic Proof

Proof. We shall first show that, in any orthonormal basis,⁷ each row $\mathbf{P}_i$ constitutes an independent optimization problem, and then select a basis in which we can easily show that the corresponding component X_i of X can be almost surely decomposed into a linear combination of mutually uncorrelated components in the whitened random vector $\mathbf{W}X$, some of which correlate with Z and some of which do not. The solution $(\mathbf{P}X)_i$ is then that same linear combination, restricted to those components which do not correlate with Z .

Consider first an orthonormal basis diagonalizing the inner product $\mathbf{M}$, so that $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{M}} = \sum_{i=1}^d \alpha_i x_i y_i$ for fixed $\alpha_1, \dots, \alpha_d \geq 0$. This allows us to treat each row $\mathbf{P}_i \in \mathbb{R}^d$ of $\mathbf{P}$ as a separate optimization problem,

$$\operatorname{argmin}_{\mathbf{P}_i \in \mathbb{R}^d} \mathbb{E} \left[\alpha_i (\mathbf{P}_i^T X - X_i)^2 \right] \quad \text{subject to } \operatorname{Cov}(\mathbf{P}_i^T X, Z) = \mathbf{0},$$

at which point the weights α_i of each subproblem become irrelevant, and our objective may as well be Euclidean, allowing us to view each row as an independent optimization problem not just in this basis, but from any convenient one.

So now let $\ell = \operatorname{rank}(\Sigma_{XZ}) = \operatorname{rank}(\Sigma_{\mathbf{W}X, Z})$ and $m = \operatorname{rank}(\Sigma_{XX}) = \operatorname{rank}(\Sigma_{\mathbf{W}X, \mathbf{W}X})$, and consider a (new) orthonormal basis whose first m coordinates span the column (and row) space of $\mathbf{W}$ (i.e. the subspace of $\mathbb{R}^d$ in which X and $\mathbf{W}X$ have nonzero variance), and whose first $\ell \leq m$ coordinates span the column space of $\Sigma_{\mathbf{W}X, Z}$ (i.e. the subspace of $\mathbb{R}^d$ in which $\mathbf{W}X$ has nonzero covariance with Z).

Any component of X can be (almost surely) written as a fixed linear combination of the nontrivial components of its whitening $\mathbf{W}X$:

$$X_i = (\mathbf{W}^+ \mathbf{W}X)_i = \sum_{j=1}^m W_{ij}^+ (\mathbf{W}X)_j. \quad (\text{almost surely})$$

Meanwhile, any component of $\mathbf{P}X$ can be (always) written as a fixed linear combination of the nontrivial components of $\mathbf{W}X$ and the almost surely zero components of X :

$$(\mathbf{P}X)_i = \sum_{j=1}^m A_{ij} (\mathbf{W}X)_j + \sum_{j=m+1}^d B_{ij} X_j,$$

i.e. $\mathbf{P} = \mathbf{A}\mathbf{W} + \mathbf{B}\mathbf{V}$, where $\mathbf{V} = \mathbf{I} - \mathbf{W}^+ \mathbf{W}$ is the orthogonal projection onto X 's almost surely zero components.

⁷Throughout this proof, we abuse the notations $X_i, \mathbf{P}_i$, etc. to refer to the i^{th} component in the specified basis, not necessarily the standard one.

The i^{th} sub-objective is then:

$$\mathbb{E}(\mathbf{P}_i^T \mathbf{X} - X_i)^2 = \mathbb{E} \left[\sum_{j=1}^m (A_{ij} - W_{ij}^+)(\mathbf{W}\mathbf{X})_j \right]^2 = \sum_{j=1}^m (A_{ij} - W_{ij}^+)^2,$$

where we have safely ignored the almost surely zero terms $B_{ij}X_j$ ($j > m$), and used the fact that the first m components of $\mathbf{W}\mathbf{X}$ have identity covariance matrix.

$\mathbf{P}\mathbf{X}$ is almost surely equal to $\mathbf{A}\mathbf{W}\mathbf{X}$, so our constraint $\text{Cov}(\mathbf{P}\mathbf{X}, \mathbf{Z}) = \mathbf{0}$ is equivalent to $\mathbf{A}\Sigma_{\mathbf{W}\mathbf{X}, \mathbf{Z}} = \text{Cov}(\mathbf{A}\mathbf{W}\mathbf{X}, \mathbf{Z}) = \mathbf{0}$, i.e. $A_{ij} = 0$ when $j \leq \ell$, since the first ℓ components are those for which $\mathbf{W}\mathbf{X}$ correlates with $\mathbf{Z}$. Subject to this, the objective is minimized for $A_{ij} = W_{ij}^+$ when $j > \ell$, i.e. $\mathbf{A} = \mathbf{W}^+(\mathbf{I} - \mathbf{P}_{\mathbf{W}\Sigma_{\mathbf{X}\mathbf{Z}}})$.

The particular choice $\mathbf{B} = \mathbf{I}$ gives our solution $\mathbf{P}^* = \mathbf{I} - \mathbf{W}^+\mathbf{P}_{\mathbf{W}\Sigma_{\mathbf{X}\mathbf{Z}}}\mathbf{W}$, leaving the non-varying components of $\mathbf{X}$ intact (see Fig. 1 for a visualization). $\square$

The solution is unique except for columns corresponding to the components of $\mathbf{X}$ with zero variance, and rows corresponding to the zero-weighted components of the (pseudo) inner product $\mathbf{M}$.

E.2 Covector Proof

Proof. We assume without loss of generality that vectors in $\mathbb{R}^d$ are represented in a basis diagonalizing the inner product $\mathbf{M}$, so that $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{M}} = \sum_{i=1}^d m_i x_i y_i$ for fixed $m_1, \dots, m_d \geq 0$. This allows us to treat each row $\mathbf{P}_i \in \mathbb{R}^d$ of $\mathbf{P}$ as a separate optimization problem,

$$\underset{\mathbf{P}_i \in \mathbb{R}^d}{\text{argmin}} \mathbb{E} \left[m_i (\mathbf{P}_i^T \mathbf{X} - X_i)^2 \right] \quad \text{subject to} \quad \text{Cov}(\mathbf{P}_i^T \mathbf{X}, \mathbf{Z}) = \mathbf{0}.$$

Our objective only depends on $\mathbf{P}_i$ through its effect on the scalar random variable $\xi = \mathbf{P}_i^T \mathbf{X}$. All random variables⁸ of the form $\zeta = \mathbf{u}_\zeta^T \mathbf{X}$ for some covector $\mathbf{u}_\zeta^T \in \mathbb{R}^d$ form a vector space U , which we equip with the covariance inner product $\langle \xi, \zeta \rangle_{\text{Cov}} = \text{Cov}(\xi, \zeta) = \mathbb{E}[\xi\zeta] = \mathbf{u}_\xi^T \Sigma_{\mathbf{X}\mathbf{X}} \mathbf{u}_\zeta$.

By the linearity of covariance, the elements of U uncorrelated with $\mathbf{Z}$ form a subspace $Z^\perp \subseteq U$. Note also that $\xi \in Z^\perp$ if and only if ξ 's covector $\mathbf{u}_\xi^T$ satisfies $\text{Cov}(\mathbf{u}_\xi^T \mathbf{X}, \mathbf{Z}) = \mathbf{u}_\xi^T \Sigma_{\mathbf{X}\mathbf{Z}} = \mathbf{0}_k$, and that these covectors themselves form the subspace $\text{colsp}(\Sigma_{\mathbf{X}\mathbf{Z}})^\perp$ of $\mathbb{R}^d$.

Our objective now reduces to finding a covector $\mathbf{P}_i^T$ that defines the orthogonal projection of X_i onto $Z^\perp$. The difficulty is that orthogonality of elements in U is not equivalent to orthogonality of the corresponding covectors. We can fix this by changing the basis in which covectors are represented. Since $\mathbf{X} \in \text{colsp}(\mathbf{W})$ a.s., we can write any element of U as a linear form in $\mathbf{W}\mathbf{X}$ rather than $\mathbf{X}$ by applying the change-of-basis $\mathbf{u}'_\xi = \mathbf{W}^+ \mathbf{u}_\xi$ to every covector: $\xi = (\mathbf{u}'_\xi)^T \mathbf{W}\mathbf{X} = \mathbf{u}'_\xi^T \mathbf{W}^+ \mathbf{W}\mathbf{X}$ a.s.

In this new basis, which is orthonormal under our covariance inner product, each component of $\mathbf{X}$ is written $X_i = (\mathbf{W}^+)_i^T \mathbf{W}\mathbf{X}$ and the inner product of any two elements of U is simply the Euclidean inner product of the corresponding covectors:⁹

$$\langle \xi, \zeta \rangle_{\text{Cov}} = \text{Cov}(\mathbf{u}'_\xi{}^T \mathbf{W}\mathbf{X}, \mathbf{u}'_\zeta{}^T \mathbf{W}\mathbf{X}) = \mathbf{u}'_\xi{}^T \mathbf{W} \Sigma_{\mathbf{X}\mathbf{X}} \mathbf{W}^+ \mathbf{u}'_\zeta = \mathbf{u}'_\xi{}^T \mathbf{u}'_\zeta.$$

Since the two inner products are now equivalent, and $Z^\perp$ is precisely those random variables with covector $\mathbf{u}' \in \text{colsp}(\mathbf{W} \Sigma_{\mathbf{X}\mathbf{Z}})^\perp$, the orthogonal projection of X_i onto $Z^\perp$ is also an orthogonal projection of its covector $(\mathbf{W}^+)_i^T$ onto $\text{colsp}(\mathbf{W} \Sigma_{\mathbf{X}\mathbf{Z}})^\perp$:

$$\hat{X}_i = (\mathbf{W}^+)_i^T (\mathbf{I} - \mathbf{P}_{\mathbf{W} \Sigma_{\mathbf{X}\mathbf{Z}}}) (\mathbf{W}\mathbf{X}) \quad (6)$$

Putting all the components of $\mathbf{X}$ together, we have our final solution,

$$\hat{\mathbf{X}} = (\mathbf{I} - \mathbf{W}^+ \mathbf{P}_{\mathbf{W} \Sigma_{\mathbf{X}\mathbf{Z}}} \mathbf{W}) \mathbf{X},$$

which is almost surely equivalent to Eq. 6, but keeps the non-varying components of $\mathbf{X}$ intact. $\square$

⁸Strictly speaking, equivalence classes of almost surely equal random variables.

⁹If $\Sigma_{\mathbf{X}\mathbf{X}}$ is full rank, there is a one-to-one correspondence between random variables in U and covectors. In the singular case, we may choose the component of the covector inside $\ker(\Sigma_{\mathbf{X}\mathbf{X}})$ arbitrarily, since it will make no difference to the inner product.